UNIVERSITY OF BIRMINGHAM

School of Physics and Astronomy

DEGREE OF B.Sc. & M.Sci. WITH HONOURS

THIRD-YEAR FINAL EXAMINATION

**03 24347**

**LH EVOLUTION OF COSMIC STRUCTURE**

**SUMMER EXAMINATIONS 2025**

***Time allowed: 1 hour 30 minutes***

Answer ***two*** questions. If you answer more than two questions, credit will only be given for the best two answers.

The approximate allocation of marks to each part of a question is shown in brackets **[ ]**.

Calculators may be used in this examination but must not be used to store text. Calculators with the ability to store text should have their memories deleted prior to the start of the examination.

A formula sheet and two tables of physical constants and units that may be required will be found at the end of this question paper.

*Answer* **two** *questions. If you answer more than two questions, credit will only be given for the best two answers.*

1. (a) Define the two-point correlation function often used in measuring galaxy clustering. Is the clustering of star-forming galaxies stronger or weaker than that of non-star forming galaxies of the same total stellar mass? Justify your answer. **[6]**

(b) Consider a spherically symmetric halo with a total mass, $M_{200}$, of $10^{14}$ $M_{\odot}$. Assume a flat universe with $H_0$ = 70 km $s^{-1}$ $Mpc^{-1}$, $\Omega_{\mathrm{M},0}$ = 0.3 and $\Omega_{\Lambda,0}$ = 0.7, where these quantities have their usual meanings.

i. Calculate $R_{200}$, the radius at which the density within it is 200 times the critical density $\rho_{\mathrm{c}}$, if this halo is at z = 0 or if this halo is at z = 1. **[8]**

ii. Show that for a halo obeying the Virial Theorem the crossing time of that halo ($t_{\mathrm{cross}}$ = $R_{200}$ / $\sigma$) at a fixed redshift does not depend on the mass of the halo. **[6]**

iii. Calculate the crossing time of the halo at z = 1 and z = 0. Comment on the effect any difference in the crossing times may have on the observable properties of galaxies in these clusters. **[6]**

iv. Ram pressure stripping of galaxies occurs when the ram pressure ($P_{\mathrm{ram}}$) is greater than the gravitational restoring force. Discuss how the ram pressure and gravitational restoring force within typical galaxies in clusters changes between z = 1 and z = 0. What effect could these changes have on the observable properties of galaxies in clusters? **[4]**

2. (a) Consider a halo with a dynamical timescale of $t_{\rm dyn}$ and a cooling timescale of $t_{\rm cool}$ at an epoch where the Hubble time is $t_H$. Describe the implications for galaxy formation if (i) $t_{\rm cool} < t_{\rm dyn} < t_H$, (ii) $t_{\rm dyn} < t_{\rm cool} < t_H$, (iii) $t_{\rm cool} >$ t$_H$. What is the typical halo mass range corresponding to each of these conditions? **[6]**

(b) Consider an N-body numerical simulation of a self-gravitating dark matter halo. The halo is a sphere of radius $R$ and contains $N$ discrete particles each with mass $m$. The particles have a uniform velocity dispersion of $\sigma_v$ in the halo.

i. Derive an expression for the relaxation time $t_r$ of this N-body halo assuming it is dominated by many small-angle, two-body interactions. The final expression should only contain $N$, $R$ and $\sigma_v$. **[12]**

ii. Consider a dark matter halo with a velocity dispersion of $\sigma_v = 300$ km/s and a system size $R = 1$ Mpc. Calculate the minimum number of particles required in a N-body simulation of this system to ensure that the dynamics are not dominated by two-body relaxation. Comment on how your estimate compares to the number of particles used in current dark matter simulations. **[6]**

(c) Describe the difference between adiabatic and isocurvature perturbations. What observations can determine which is the dominant mode for the initial density perturbations in our Universe? **[6]**

3. (a) Describe, with the aid of a diagram, how the shape of the matter power spectrum $P(k)$ evolves during the radiation-dominated epoch of our Universe, where $k$ represents the comoving wavenumber. **[6]**

(b) The $y$-parameter measures the energy transfer from hot gas in galaxy clusters to Cosmic Microwave Background photons through inverse Compton scattering, a process known as the thermal Sunyaev-Zeldovich (tSZ) effect. The $y$-parameter along a line of sight $l$ is given by

$$y = \int \frac{k_B T_e}{m_e c^2} n_e \sigma_T \, dl \ ,$$

where $k_B$ is the Boltzmann constant, $T_e$ is the electron temperature, $m_e$ is the electron mass, $c$ is the speed of light, $n_e$ is the electron number density, and $\sigma_T$ is the Thomson cross-section.

i. Derive an expression for $y$ assuming an isothermal sphere with an electron density profile of a galaxy cluster of the form

$$n_e(r) = n_{e,0} \left(1 + \frac{r^2}{r_c^2}\right)^{-3/2} \ ,$$

where $r_c$ is a characteristic core radius. **[6]**

ii. Use the Virial Theorem to find an expression for the temperature of the cluster gas as a function of the cluster total mass $M_{200}$. **[6]**

iii. Calculate the total tSZ flux $Y$ (the integrated $y$-parameter) as a function of cluster total mass $M_{200}$. **[8]**

iv. Describe how the tSZ scaling relation can be used to constrain the matter density $\Omega_m$ and the amplitude of fluctuations on an 8 Mpc scale $\sigma_8$. **[4]**

# Formula Sheet

Friedmann equation
$H^2 = \frac{8}{3}\pi G\rho - \frac{k}{a^2}$

Fluid equation
$\dot{\rho} + 3\frac{\dot{a}}{a}\left(\rho + P\right) = 0$

Acceleration equation
$\frac{\ddot{a}}{a} = -\frac{4}{3}\pi G\left(\rho + 3P\right)$

Equation of State
$P = \omega\rho$

Critical density
$\rho_c = \frac{3H^2}{8\pi G}$

Hubble parameter
$H = \frac{\dot{a}}{a}$

$H(z) = H_0\sqrt{\frac{\Omega_{M,0}}{a^3} + \frac{\Omega_{r,0}}{a^4} + \Omega_{\lambda,0}}$

Scale factor (Matter only)
$a(t) \propto t^{2/3}$

Scale factor (Radiation only)
$a(t) \propto t^{1/2}$

Overdensity
$\delta(r,t) = \frac{\rho(r,t) - \bar{\rho}(t)}{\bar{\rho}(t)}$

Virial theorem
$2T + V = 0$

Dynamical time (gravitational)
$t_{dyn} = \frac{1}{\sqrt{4\pi G\bar{\rho}}}$

Press-Schechter number density
$\frac{dn}{d\log M} = \sqrt{\frac{2}{\pi}}\frac{\rho_{\rm M}}{M}\left|\frac{n_{\rm eff}+3}{6}\right|\nu e^{-\nu^2/2}$

NFW density profile
$\rho(r) = \frac{\rho_0}{(r/r_s)(1+r/r_s)^2}$

Optical depth
$\tau_e = \int n_e \sigma_T dx$

Feedback energy from supernova
$E_{fb} = \epsilon_{SN}\,\xi\,M_*\,E_{SN}$

Feedback energy from black holes
$E_{AGN} = \epsilon_{BH}\,M_{BH}\,c^2$

Radiation pressure
$P_{rad} = \frac{L}{c}\frac{1}{4\pi R^2}$

Ram pressure condition
$\rho_{icm}v^2 = 2\pi G\Sigma(R)\Sigma_g(R)$

X-ray luminosity
$L_x \propto \rho^2\,V\Lambda(T)$

Correlation function
$\langle N_g\rangle = n[1 + \xi(r)]V$

Radius of spherical top hat model
$R(t) = \frac{R_*}{2}\left(\frac{6t}{\alpha R_*}\right)^{2/3}\left[1 - \frac{1}{20}\left(\frac{6t}{\alpha R_*}\right)^{2/3}\right]$

# Astrophysical Constants and Units

| | | |
|---|---|---|
| Astronomical Unit | AU | $1.50 \times 10^{11}$ m |
| Parsec | pc | $3.1 \times 10^{16}$ m |
| Tropical year | y | $365.242$ mean solar days |
| Solar luminosity | $L_\odot$ | $3.84 \times 10^{26}$ W |
| Absolute bolometric magnitude of the Sun | $M_{\rm bol}$ | $+4^{\rm m}75$ |
| Bolometric correction for the Sun | BC | $-0^{\rm m}08$ |
| Apparent visual magnitude of the Sun | $m_v(\odot)$ | $-26^{\rm m}74$ |
| Solar constant | $f$ | $1.36 \times 10^3$ W m$^{-2}$ |
| Solar mass | $M_\odot$ | $1.989 \times 10^{30}$ kg |
| Solar radius | $R_\odot$ | $6.95 \times 10^8$ m |
| Solar effective temperature | $T_{\rm eff}$ | 5777 K |
| Wien constant | $b$ | $2.898 \times 10^{-3}$ m K |
| Hubble constant | $H_o$ | 70 km s$^{-1}$ Mpc$^{-1}$ |
| Distance of the Sun from galactic centre | | 8.3 kpc |
| Earth mass | $M_\oplus$ | $5.972 \times 10^{24}$ kg |
| Earth radius | $R_\oplus$ | 6371 km |
| Jupiter mass | $M_{\rm Jup}$ | $1.898 \times 10^{27}$ kg |
| Jupiter radius | $R_{\rm Jup}$ | 71 492 km |

# Physical Constants and Units

| | | |
|---|---|---|
| Acceleration due to gravity | $g$ | $9.81\,\mathrm{m\,s^{-2}}$ |
| Gravitational constant | $G$ | $6.674 \times 10^{-11}\,\mathrm{N\,m^2\,kg^{-2}}$ |
| Ice point | $T_{ice}$ | $273.15\,\mathrm{K}$ |
| Avogadro constant | $N_A$ | $6.022 \times 10^{23}\,\mathrm{mol^{-1}}$ |
| | | [*N.B.* 1 mole $\equiv 1$ *gram-molecule*] |
| Gas constant | $R$ | $8.314\,\mathrm{J\,K^{-1}\,mol^{-1}}$ |
| Boltzmann constant | $k, k_B$ | $1.381 \times 10^{-23}\,\mathrm{J\,K^{-1}} \equiv 8.62 \times 10^{-5}\,\mathrm{eV\,K^{-1}}$ |
| Stefan constant | $\sigma$ | $5.670 \times 10^{-8}\,\mathrm{W\,m^{-2}\,K^{-4}}$ |
| Rydberg constant | $R_\infty$ | $1.097 \times 10^{7}\,\mathrm{m^{-1}}$ |
| | $R_\infty hc$ | $13.606\,\mathrm{eV}$ |
| Planck constant | $h$ | $6.626 \times 10^{-34}\,\mathrm{J\,s} \equiv 4.136 \times 10^{-15}\,\mathrm{eV\,s}$ |
| $h/2\pi$ | $\hbar$ | $1.055 \times 10^{-34}\,\mathrm{J\,s} \equiv 6.582 \times 10^{-16}\,\mathrm{eV\,s}$ |
| Speed of light *in vacuo* | $c$ | $2.998 \times 10^{8}\,\mathrm{m\,s^{-1}}$ |
| | $\hbar c$ | $197.3\,\mathrm{MeV\,fm}$ |
| Charge of proton | $e$ | $1.602 \times 10^{-19}\,\mathrm{C}$ |
| Mass of electron | $m_e$ | $9.109 \times 10^{-31}\,\mathrm{kg}$ |
| Rest energy of electron | | $0.511\,\mathrm{MeV}$ |
| Mass of proton | $m_p$ | $1.673 \times 10^{-27}\,\mathrm{kg}$ |
| Rest energy of proton | | $938.3\,\mathrm{MeV}$ |
| One atomic mass unit | $u$ | $1.66 \times 10^{-27}\,\mathrm{kg}$ |
| Atomic mass unit energy equivalent | | $931.5\,\mathrm{MeV}$ |
| Electric constant | $\epsilon_0$ | $8.854 \times 10^{-12}\,\mathrm{F\,m^{-1}}$ |
| Magnetic constant | $\mu_0$ | $4\pi \times 10^{-7}\,\mathrm{H\,m^{-1}}$ |
| Bohr magneton | $\mu_B$ | $9.274 \times 10^{-24}\,\mathrm{A\,m^2}$ ($\mathrm{J\,T^{-1}}$) |
| Nuclear magneton | $\mu_N$ | $5.051 \times 10^{-27}\,\mathrm{A\,m^2}$ ($\mathrm{J\,T^{-1}}$) |
| Fine-structure constant | $\alpha = e^2/4\pi\epsilon_0\hbar c$ | $7.297 \times 10^{-3}$= 1/137.0 |
| Compton wavelength of electron | $\lambda_c = h/m_e c$ | $2.426 \times 10^{-12}\,\mathrm{m}$ |
| Bohr radius | $a_0$ | $5.2918 \times 10^{-11}\,\mathrm{m}$ |
| angstrom | Å | $10^{-10}\,\mathrm{m}$ |
| barn | b | $10^{-28}\,\mathrm{m^2}$ |
| torr (mm Hg at $0\,^\circ$C) | torr | $133.32\,\mathrm{Pa}$ ($\mathrm{N\,m^{-2}}$) |

**Do not complete the attendance slip, fill in the front of the answer book or turn over the question paper until you are told to do so.**

**Important Reminders**

- Coats/outwear should be placed in the designated area.
- Unauthorised materials (e.g. notes or Tippex) must be placed in the designated area.
- Check that you do not have any unauthorised materials with you (e.g. in your pockets, pencil case).
- Mobile phones and smart watches **must** be switched off and placed in the designated area or under your desk. They must not be left on your person or in your pockets.
- You are not permitted to use a mobile phone as a clock. If you have difficulty seeing a clock, please alert an Invigilator.
- You are not permitted to have writing on your hand, arm or other body part.
- Check that you do not have writing on your hand, arm or other body part – if you do, you must inform an Invigilator immediately
- Alert an Invigilator immediately if you find any unauthorised item upon you during the examination.

**Any students found with non-permitted items upon their person during the examination, or who fail to comply with Examination rules may be subject to Student Conduct procedures.**